ELS_Interface — Hardware Configuration

This document explains **how the board is wired**: which microcontroller pin goes where, which connectors carry the signals to the outside world and at which voltages everything works.

It contains no software: it describes only the physical connection between the components. It is meant for whoever writes the program (to know which pin to use), for whoever connects external equipment, and for whoever has to find a fault.

Board: NUCLEO-G474RE with the **STM32G474RET6** microcontroller (LQFP64 package). The shield sits on the two morpho headers.

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1. How to read this document

Before the tables, three names that come up all the time:

Term	Meaning	Example
PAn / PBn / PCn	name of a microcontroller pin	**PA0** = pin 0 of port A
J1, J2, J3, J5, J7, J8	name of a connector on the board	**J5** = the 2x10 field connector
Net	name given to an electrical connection	**E5V** = the 5 V line of the converter

When a table says "PA0 → J1-28" it means: pin PA0 of the microcontroller is connected to **pin 28 of connector J1**, so the signal is also available at the connector.

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2. Pins and peripherals

2.1 Digital outputs

MCU pin	Connected to	What it does	Note
PA0	R19 (47 Ohm) → LED1	lights the green LED	also available on J1-28
PA1	R18 (100 Ohm) → LED2	lights the red LED	also available on J1-30

MCU pin	Connected to	What it does	Note
PA5	LD2 LED of the Nucleo	LED of the Nucleo board	no component on the shield
PA8	connector J1	free output	available on the morpho
PA9	connector J1	free output	available on the morpho

All of these work at **3.3 V** and can source about 8 mA each.

Practical example: to light the green LED just drive PA0 high; the LED stays off while PA0 is low.

2.2 Serial lines (towards the display)

MCU pin	Signal	Connected to	Level	Note
PB6	DWIN_TX	J7 pin 5	5 V	transmits towards the display
PA10	DWIN_RX	J7 pin 4	5 V	receives from the display through **R4 (100 Ohm)**
PA2	—	Nucleo board	—	Nucleo debug console
PA3	—	Nucleo board	—	Nucleo debug console

PA10 and PB6 are **5 V tolerant** pins: the display can drive them directly, with no level shifter. R4 is essential: it limits the current spikes along the cable.

2.3 EEPROM memory (I2C bus)

Signal	MCU pin	Memory pin	Components	Voltage
SCL (clock)	**PC8**	6	**R16 = 4.7 kOhm** to 3.3 V	3.3 V
SDA (data)	**PC11**	5	**R17 = 4.7 kOhm** to 3.3 V	3.3 V

Signal	MCU pin	Memory pin	Components	Voltage
WC (write protect)	—	7	connected to ground	writing always allowed
Supply	—	8	3.3 V + **C14 (100 nF)**	3.3 V
Address A0-A1-A2	—	—	all to ground	address **0x50**

The memory is an **M24M01E** of 1 Mbit: it keeps its data when the board is powered off (for example machine parameters or positions).

> **Confirmed:** the **PC8/PC11** pair is the microcontroller **I2C3** — PC8 = `I2C3_SCL`, PC11 = `I2C3_SDA` (alternate function table of the STM32G474 datasheet). The bus is therefore used as hardware I2C, with no bit-banging.

2.4 Encoders

Signal	MCU pin	Channel	Connector	Note
JOG_ENC_A	**PA6**	TIM3_CH1	J5-8	manual (jog) encoder
JOG_ENC_B	**PA4**	TIM3_CH2	J5-9	second channel of the same encoder
PC0	**PC0**	TIM1_CH1	J5-3	spindle encoder, channel A
SPINDLE_ENC_B	**PC1**	TIM1_CH2	J5-4	spindle encoder, channel B

Encoder signals are **two square waves 90 degrees apart**: from the sequence the microcontroller knows how far and in which direction the axis is turning. Pins PA6, PA4 and PC1 are not 5 V tolerant: they reach J5 through the level shifter **U2**.

2.5 Buttons and emergency stop

Signal	MCU pin	Connector	Components	Behaviour
BTN_JOG_PLUS	**PB14**	J5-15	internal pull-up	**active low**
BTN_JOG_MINUS	**PA15**	J5-16	internal pull-up	**active low**
BTN_MODE	**PB12**	J5-14	internal pull-up	**active low**
BTN_RESET	**PB10**	J5-13	internal pull-up	**active low**
EMERGENCY	**PB7**	J5-11	**R3 = 1 kOhm** in series + **C7 = 100 nF** to GND (RC filter, ~100 us)	**active low**

What "active low" means: at rest the pin sits at 3.3 V; when you press the button the pin drops to 0 V. A button wired to ground is therefore normal.

> **Warning: PB7 is dedicated to the emergency stop** and must not be reassigned: it is the pin that guarantees the immediate stop.

2.6 Pins reserved for programming

Pin	Signal	Note
PA13	SWDIO	reserved for programming, do not use for anything else
PA14	SWCLK	reserved for programming
PB3	SWO	reserved for the debug trace

2.7 Supplies available on the connectors

Rail	Connector	Voltage	Max current
E5V	J1-6	5.00 V	**500 mA** (Nucleo limit)
V_5V_SHIELD	J5-1 and J5-7	4.4–4.7 V	about 100 mA in total

Rail	Connector	Voltage	Max current
Display 12 V	J7-1 and J7-2	12 V	0.5 A (F2 limit)
3.3 V	J1-12 and J1-16	3.30 V	supplied by the Nucleo

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3. Hardware to connector map

The master table, to use when you need to find "where this signal ends up".

Signal	MCU pin	Components on the board	Connector	Voltage
+12 V input	—	F1, D1, C1, C3	J3-1	12 V
Ground	—	—	J3-2, J8-5, J7-7/8	0 V
E5V	—	L1, C5, C15, D4 anode	**J1-6**	5.00 V
V_5V_SHIELD	—	D4, D5, C12	J5-1, J5-7	4.4–4.7 V
USB VBUS	—	D5 anode	J2-8	5 V
Display 12 V	—	F2, D3, C6	J7-1, J7-2	12 V
Green LED	**PA0**	R19 47 Ohm → LED1	J1-28	3.3 V
Red LED	**PA1**	R18 100 Ohm → LED2	J1-30	3.3 V
DWIN_TX	**PB6**	—	J7-5	5 V
DWIN_RX	**PA10**	R4 100 Ohm	J7-4	5 V
Memory SCL	**PC8**	R16 4.7 kOhm, C14	J2-2	3.3 V
Memory SDA	**PC11**	R17 4.7 kOhm	J1-2	3.3 V
Jog encoder A	**PA6**	U2 (A1→B1)	J5-8	3.3 → 4.6 V
Jog encoder B	**PA4**	U2 (A2→B2)	J5-9	3.3 → 4.6 V
Spindle encoder A	**PC0**	—	J5-3	3.3 V

Signal	MCU pin	Components on the board	Connector	Voltage
Spindle encoder B	**PC1**	U2 (A3→B3)	J5-4	3.3 → 4.6 V
Auxiliary signal	**PC4**	—	J5-5	3.3 V
Emergency	**PB7**	R3 1 kOhm in series + C7 100 nF	J5-11	3.3 V, active low
Reset button	**PB10**	internal pull-up	J5-13	active low
Mode button	**PB12**	internal pull-up	J5-14	active low
Jog + button	**PB14**	internal pull-up	J5-15	active low
Jog − button	**PA15**	internal pull-up	J5-16	active low
Isolated output 1	via U3	R12 220 Ohm, PC817	J8-1	open collector
Isolated output 2	via U4	R13 220 Ohm, PC817	J8-2	open collector
Isolated output 3	via U5	R14 220 Ohm, PC817	J8-3	open collector
Direct output	**PC9**	R15 4.7 kOhm (pull-up)	J8-4	3.3 V
+3.3 V	—	R16, R17, C13, C14	J1-12, J1-16	3.3 V

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4. Logic levels

The board works with **three different voltages**, and this is where most mistakes happen. Every "family" of signals is separated from the others.

Domain	Voltage	What lives there
Microcontroller	**3.3 V**	Nucleo, EEPROM, I2C pull-ups, LEDs, opto series resistors
Board (internal rail)	**4.4–5.0 V**	supply of shifter U2, opto collectors, 5 V pins of J5

Domain	Voltage	What lives there
Field (external)	**5 V** supplied by you	buttons, encoders, output loads
Display	**12 V** power + **5 V** signals	connector J7

The bridge between 3.3 V and 5 V is the level shifter **U2 (TXB0104)**: it takes a signal on one side and repeats it on the other with the right voltage. Its main pins:

U2 pin	Function	Connected to
1 and 8	VCCA (3.3 V side)	3.3 V + C13 (100 nF)
14	VCCB (5 V side)	V_5V_SHIELD + C12 (100 nF)
2, 3, 4	inputs A1, A2, A3	PA4, PA6, PC1
11, 12, 13	outputs B1, B2, B3	J5 pins

> **Rule of thumb:** a signal going from the microcontroller (3.3 V) to the outside (5 V) **must** pass through U2. A 5 V signal entering a non-tolerant pin damages it.

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5. Design notes

#	Note
N 1	**R1 and R2** (10 kOhm and 3.24 kOhm) set the converter output to 5.00 V. If you change them, E5V changes too and the 4.75–5.25 V window required by the Nucleo must be re-verified
N 2	**D4 and D5 are in SOD-123** packages (pads 0.9 x 1.2 mm). A SOD-123FL part does not fit: it is not a compatible substitute
N 3	The thermal vias under **U1** have a 0.150 mm annular ring, against the 0.180 mm recommended by JLCPCB for manufacturing. They are still above the 0.130 mm absolute minimum, but they are the most critical point of the board
N 4	The **V_5V_SHIELD** rail always sits between 4.4 and 4.7 V because of the diode drop: devices connected to J5 must accept this voltage

#	Note
N 5	The **PC8/PC11** pair is **confirmed as I2C3**: PC8 = `I2C3_SCL`, PC11 = `I2C3_SDA` (alternate function table of the STM32G474 datasheet). **PB7** (emergency) is also an I2C pin (`I2C1_SDA`, `I2C4_SDA`) and must stay configured as GPIO
N 6	**3 excluded DRC findings** (project-level *Local override*): 1 **error** `footprint_type_mismatch` on **U1** (footprint declares through-hole but its pads are SMD) and 2 **warnings** `lib_footprint_mismatch` on **F1** and **U1** (locally modified footprint differs from the library copy). They are intentional and reviewed: they do not block manufacturing

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6. Design verification

The configuration described here was checked with the **`kicad-mcp-pro`** MCP server (KiCad 10.0.6) and with `kicad-cli`:

Check	Result
ERC — electrical rules check of the schematic	**0 violations**
DRC — design rules check of the PCB	**0 active violations**, **0 unconnected items**, **3 excluded findings** (note **N6**)
Netlist	exported correctly, **56 components**
Schematic / PCB parity	**56 = 56**, no missing component
3D models	**56 of 56** resolved
DFM (JLCPCB)	2 layers, minimum track 0.25 mm, via 0.40/0.80 mm